# 4 Systems of First Order Linear Ordinary Differential Equations

*Reference: Boyce & DiPrima, Chapter 6*

Systems of ordinary differential equations arise when there is a single independent variable, $t$, say, and $n$ dependent variables $x_1(t), x_2(t), \ldots x_n(t)$ (the unknowns), each of which is a function of $t$. Physically, we are are modelling problems in which there are a number of separate elements that are linked together in some way. Mathematically, this structure is very useful because any higher-order ODE can be reformulated as a system of first-order ODEs.

## 4.1 General framework

The most general system of $n$ ordinary differential equations for $x_1(t), x_2(t), \ldots x_n(t)$ has the form

$$\begin{aligned} \frac{dx_1}{dt} &= f_1(x_1, x_2, \ldots, x_n)\,, \\ \frac{dx_2}{dt} &= f_2(x_1, x_2, \ldots, x_n)\,, \\ &\vdots \\ \frac{dx_n}{dt} &= f_n(x_1, x_2, \ldots, x_n)\,. \end{aligned} \tag{4.1}$$

The system (4.1) has a **solution** on the interval $I : \alpha < t < \beta$ if there exists a set of $n$ functions

$$x_1(t), \quad x_2(t), \quad \ldots, \quad x_n(t) \tag{4.2}$$

that are differentiable at all points in the interval $I$ and that satisfy the equations (4.1) at all points in this interval. We might also be given **initial conditions**

$$x_1(t_0) = x_1^0, \quad x_2(t_0) = x_2^0, \quad \ldots \quad x_n(t_0) = x_n^0\,, \tag{4.3}$$

where $t_0$ is a specified value of $t$ in the interval $I$ and the $x_1^0, x_2^0, \ldots, x_n^0$ are given numbers. The differential equations (4.1) and the initial conditions (4.3) together form an **initial value problem**.

The solution (4.2) can be viewed as a set of parametric equations in an $n$-dimensional space: for a given value of the parameter $t$, equations (4.2) tell us the coordinates $x_1, x_2, \ldots, x_n$ of a point in space. As $t$ changes, the coordinates change and the collection of points for $\alpha < t < \beta$ form a curve. We can think of the curve as the path (or trajectory) of a particle moving according to the system of equations (4.1). The initial conditions (4.3) determine the starting point of the moving particle.

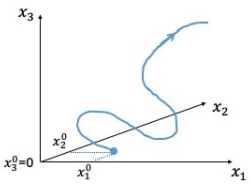



Figure 4.1: A particle that at $t = t_0$ is located at position $(x_1^0, x_2^0, x_3^0)$ moves along the path described by the parametric curve (4.2).

In vector notation, the system of first order ODEs (4.1) can be written as

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{f}(\boldsymbol{x}, t)\,, \tag{4.4}$$

where

$$\boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad \boldsymbol{f}(\boldsymbol{x}, t) = \begin{pmatrix} f_1(\boldsymbol{x}, t) \\ f_2(\boldsymbol{x}, t) \\ \vdots \\ f_n(\boldsymbol{x}, t) \end{pmatrix}, \tag{4.5}$$

i.e. $\boldsymbol{x}$ is a column vector (or $n \times 1$ matrix) whose elements are the unknown functions of $t$, and $\boldsymbol{f}$ is a column vector of the right-hand sides of (4.1). The initial conditions (4.3) are written as

$$\boldsymbol{x}(t_0) = \boldsymbol{k}\,. \tag{4.6}$$

It is apparent that we need to deal with **matrix functions**, i.e. matrices whose elements are functions of $t$.

**Example 4.1** Consider a particle that moves on a curve defined by

$$\boldsymbol{x}(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}.$$

Find $\frac{d\boldsymbol{x}}{dt}$ and interpret your result geometrically.

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## 4.2 Revision: Matrices and linear algebraic systems

In order to be able to deal with systems of differential equations, students are strongly encouraged to revise the material in MTH1002 on matrices and linear algebraic systems. In particular, students should be comfortable with the material summarised in Appendices A and B (see handouts).

## 4.3 Basic theory of systems of first order linear ODEs

**Definition 4.2** If each of the functions $f_1, \ldots, f_n$ in system (4.1) is a linear function of the dependent variables $x_1, \ldots, x_n$, then the system of equations is said to be **linear**. Otherwise it is **nonlinear**.

The most general form of $n$ first order linear ordinary differential equations is

$$\begin{aligned} \frac{dx_1}{dt} &= p_{11}(t)x_1 + \cdots + p_{1n}(t)x_n + g_1(t)\,, \\ \frac{dx_2}{dt} &= p_{21}(t)x_1 + \cdots + p_{2n}(t)x_n + g_2(t)\,, \\ &\vdots \\ \frac{dx_n}{dt} &= p_{n1}(t)x_1 + \cdots + p_{nn}(t)x_n + g_n(t)\,. \end{aligned} \tag{4.7}$$

We write the system (4.7) in matrix form as

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} + \boldsymbol{g}(t)\,. \tag{4.8}$$

The $n \times n$ matrix

$$\boldsymbol{P} = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \ldots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \ldots & p_{2n}(t) \\ \vdots & \vdots & \ldots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \ldots & p_{nn}(t) \end{pmatrix}$$

is called the matrix of coefficients and the column vector

$$\boldsymbol{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix}$$

is the vector of free terms. The solution is

$$\boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} \tag{4.9}$$

and initial conditions (4.3) or (4.6) are

$$\boldsymbol{x}(t_0) = \boldsymbol{k}\,. \tag{4.10}$$

**Theorem 4.3** If $\boldsymbol{P}$ and $\boldsymbol{g}$ are continuous on an open interval $I : \alpha < t < \beta$ (i.e. the functions $p_{11}, p_{12}, \ldots, p_{nn}, g_1, \ldots, g_n$ are continuous on $I$), with $t_0 \in I$, then there exists a **unique solution** of the linear system (4.8) that satisfies the initial conditions (4.10). The solution exists throughout the interval $I$.

**Definition 4.4** A **homogeneous** system of linear ODEs is a linear system in which $\boldsymbol{g}(t) = \mathbf{0}$ for all $t$ in the interval $I$. Hence (4.8) becomes

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x}. \tag{4.11}$$

As before, we'll see that once the homogeneous system has been solved there are methods to solve the **nonhomogeneous** ($\boldsymbol{g} \neq \mathbf{0}$) system (see §4.7).

**Theorem 4.5 — Principle of superposition for homogeneous linear systems.** If the vector functions $\boldsymbol{x}^{(1)}(t)$ and $\boldsymbol{x}^{(2)}(t)$ are solutions of the homogeneous system (4.11) then the linear combination

$$\boldsymbol{x}(t) = c_1\boldsymbol{x}^{(1)}(t) + c_2\boldsymbol{x}^{(2)}(t)$$

is also a solution for any constants $c_1$ and $c_2$.

We'll use the notation

$$\boldsymbol{x}^{(1)}(t) = \begin{pmatrix} x_{11}(t) \\ x_{21}(t) \\ \vdots \\ x_{n1}(t) \end{pmatrix}, \quad \ldots, \quad \boldsymbol{x}^{(k)}(t) = \begin{pmatrix} x_{1k}(t) \\ x_{2k}(t) \\ \vdots \\ x_{nk}(t) \end{pmatrix}, \ldots \tag{4.12}$$

to denote the specific solutions of (4.11), where $x_{ij}(t) = x_i^{(j)}(t)$ refers to the $i^{th}$ component of the $j^{th}$ solution $\boldsymbol{x}^{(j)}(t)$.

We see that $\boldsymbol{x}(t) = \mathbf{0}$ is always a solution to (4.11). We call it the **trivial solution**.

**Example 4.6** Write the linear system

$$\frac{dx_1}{dt} = x_1(t) + x_2(t),$$
$$\frac{dx_2}{dt} = 4x_1(t) + x_2(t),$$

in the matrix form (4.11). Then verify that

$$\boldsymbol{x}^{(1)}(t) = \begin{pmatrix} e^{3t} \\ 2e^{3t} \end{pmatrix}, \quad \boldsymbol{x}^{(2)}(t) = \begin{pmatrix} e^{-t} \\ -2e^{-t} \end{pmatrix}$$

are solutions of (4.11). Plot the vector function $\boldsymbol{x}^{(1)}(t)$.

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**Activity 4.7** According to Theorem 4.5

$$\boldsymbol{x} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$$

also satisfies (4.11), for arbitrary constants $c_1$ and $c_2$. Verify this.

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**Definition 4.8** The determinant

$$W[\boldsymbol{x}^{(1)},\ldots,\boldsymbol{x}^{(n)}] = \left|\boldsymbol{x}^{(1)}\,\boldsymbol{x}^{(2)}\,\ldots\,\boldsymbol{x}^{(n)}\right| = \begin{vmatrix} x_{11}(t) & \ldots & x_{1n}(t) \\ \vdots & & \vdots \\ x_{n1}(t) & \ldots & x_{nn}(t) \end{vmatrix} \tag{4.13}$$

is called the **Wronskian of the $n$ solutions** $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ of system (4.11).

**Theorem 4.9** If $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ are solutions of system (4.11) on the interval $I : \alpha < t < \beta$, then in this interval the Wronskian $W[\boldsymbol{x}^{(1)},\ldots,\boldsymbol{x}^{(n)}]$ either is identically zero or it never vanishes. The solutions are **linearly independent** on $I$ if and only if

$$W[\boldsymbol{x}^{(1)},\ldots,\boldsymbol{x}^{(n)}] \neq 0\,.$$

**Theorem 4.10** If $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ are linearly independent solutions of system (4.11) on the interval $I : \alpha < t < \beta$, then each solution $\boldsymbol{x}(t)$ of that system can be expressed as a linear combination

$$\boldsymbol{x}(t) = c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_n\boldsymbol{x}^{(n)}(t) \tag{4.14}$$

in exactly one way. We call (4.14) the **general solution** of (4.11). The set of solutions $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ is said to be a **fundamental set of solutions** for the interval $I$.

## 4.4 Homogeneous linear systems with constant coefficients

We consider the simplest case

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x}\,, \tag{4.15}$$

where $\boldsymbol{A}$ is a constant $n \times n$ matrix. This is sometimes called an **autonomous ODE system** because $\boldsymbol{A}$ is not a function of the independent variable $t$.

If $n = 1$ we have a single first order differential equation

$$\frac{dx}{dt} = ax, \quad a \neq 0 \tag{4.16}$$

with solution $x = ce^{at}$. The **equilibrium solution**[11] is defined by $dx/dt = 0$ and hence the only equilibrium solution is $x = 0$. Figures 4.2 and 4.3 show the solutions in the $xt$-plane (the integral curves). If $a > 0$, solutions move away from $x = 0$. If $a < 0$, solutions move towards the equilibrium solution.

[11] aka equilibrium point, critical point or fixed point

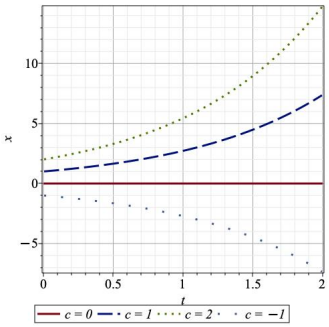



Figure 4.2: The solutions to (4.16), namely $x(t) = ce^{at}$ for different $c$ with $a = 1 > 0$, move away from the equilibrium $x = 0$.

For higher order systems we follow a similar method: Equilibrium solutions of (4.15) are found by solving

$$A\boldsymbol{x} = \boldsymbol{0}\,.$$

If $|\boldsymbol{A}| \neq 0$ then $\boldsymbol{x} = \boldsymbol{0}$ is the only equilibrium solution. It is of interest whether other solutions approach the equilibrium or move away from it as $t$ increases.

Visualisation is easiest if we take the case $n = 2$. We let $\boldsymbol{x} = \boldsymbol{x}(t)$ be a solution of (4.15) for $t \in I$, with the initial condition $\boldsymbol{x}(t_0) \neq \boldsymbol{0}$, $t_0 \in I$. Then, as $t$ varies, $\boldsymbol{x}(t) = (x_1(t), x_2(t))$ describes a parametric curve in the $x_1x_2$-plane with $t$ as the parameter. The $x_1x_2$-plane is referred to as the **phase space** (or phase plane for the two-dimensional case). The **phase portrait** of system (4.15) is a phase-space plot of a representative set of solution curves (sometimes called trajectories) $\boldsymbol{x}(t)$ corresponding to different $\boldsymbol{x}(t_0)$. An example of a phase portrait is shown in figure 4.5.

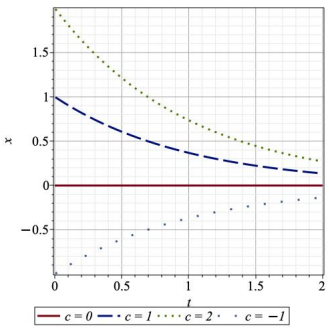



Figure 4.3: As for figure 4.2 except $a = -1 < 0$, showing that solutions approach the equilibrium.

To find the general solution of (4.15) we proceed in a similar way as we did for second order linear ODEs with constant coefficients, i.e. we seek solutions in the form of an exponential

$$\boldsymbol{x}(t) = \boldsymbol{\xi}e^{rt}\,, \tag{4.17}$$

where the (constant) exponent $r$ and the (constant) vector $\boldsymbol{\xi}$ are to be determined. Substituting (4.17) into (4.15) gives

$$r\boldsymbol{\xi}e^{rt} = \boldsymbol{A}\boldsymbol{\xi}e^{rt}\,.$$

Hence we require that

$$\boldsymbol{A}\boldsymbol{\xi} = r\boldsymbol{\xi}\,, \tag{4.18}$$

i.e. the vector $\boldsymbol{x}$ given by (4.17) is a solution of the system (4.15) provided that $r$ is an eigenvalue and $\boldsymbol{\xi}$ is the corresponding eigenvector of the matrix $\boldsymbol{A}$.

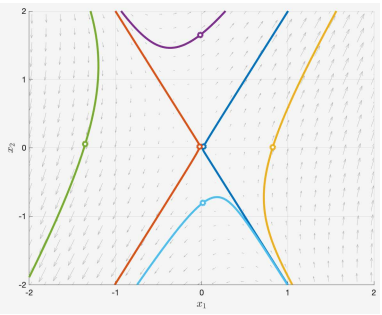


Figure 4.4: An example of a phase portrait.

To find the eigenvalues we must find the roots of the characteristic equation

$$|\boldsymbol{A} - r\boldsymbol{I}| = 0\,,$$

which is a polynomial of degree $n$. Similar to the case for second order ODEs, when $n = 2$ the roots can be either:

- Real and different
- Complex conjugates
- Real and repeated

We know by Theorem 4.10 that the general solution of an $n \times n$ linear system is a linear combination of $n$ linearly independent solutions. Hence, if the matrix $\boldsymbol{A}$ has $n$ linearly independent eigenvectors, $\boldsymbol{\xi}^{(j)}$ for $j = 1, \ldots, n$, with corresponding eigenvalues $r_j$, then the general solution of the homogeneous system (4.15) is

$$\boldsymbol{x}(t) = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + \cdots + c_n\boldsymbol{\xi}^{(n)}e^{r_n t}\,. \tag{4.19}$$

### 4.4.1 Real and different eigenvalues

**Example 4.11** Find the general solution of the autonomous system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}\boldsymbol{x}$$

and sketch the phase portrait.

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**Activity 4.12** Find the general solution of the autonomous system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -3 & \sqrt{2} \\ \sqrt{2} & -2 \end{pmatrix} \boldsymbol{x}$$

and sketch the phase portrait.

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### 4.4.2 Complex conjugate eigenvalues

For a real matrix $\boldsymbol{A}$, complex eigenvalues appear in complex conjugate pairs. For example, if $r_1 = \lambda + i\mu$ (where $\lambda$ and $\mu$ are real) is an eigenvalue then so is $r_2 = \lambda - i\mu$. Furthermore, the corresponding eigenvectors $\boldsymbol{\xi}^{(1)}$ and $\boldsymbol{\xi}^{(2)}$ are also complex conjugates. This can be shown by considering the equation for $r_1$ and $\boldsymbol{\xi}^{(1)}$

$$(\boldsymbol{A} - r_1\boldsymbol{I})\boldsymbol{\xi}^{(1)} = \boldsymbol{0} \tag{4.20}$$

and taking the complex conjugate to obtain

$$(\boldsymbol{A} - \bar{r_1}\boldsymbol{I})\bar{\boldsymbol{\xi}}^{(1)} = \boldsymbol{0}\,, \tag{4.21}$$

since $\boldsymbol{A}$ and $\boldsymbol{I}$ are real. Hence $r_2 = \bar{r_1}$ is an eigenvalue with eigenvector $\boldsymbol{\xi}^{(2)} = \bar{\boldsymbol{\xi}}^{(1)}$. Thus

$$\boldsymbol{x}^{(1)}(t) = \boldsymbol{\xi}^{(1)}e^{r_1 t}, \quad \boldsymbol{x}^{(2)}(t) = \bar{\boldsymbol{\xi}}^{(1)}e^{\bar{r_1}t}, \tag{4.22}$$

are solutions of (4.15) and are complex conjugates of each other.

As before, we can find two real-valued solutions corresponding to the eigenvalues $r_1$ and $r_2$ by taking the real and imaginary parts of $\boldsymbol{x}^{(1)}$ (or $\boldsymbol{x}^{(2)}$). If we write $\boldsymbol{\xi}^{(1)} = \boldsymbol{a} + i\boldsymbol{b}$, where $\boldsymbol{a}$ and $\boldsymbol{b}$ are real, then we have

$$\begin{aligned}
\boldsymbol{x}^{(1)}(t) &= (\boldsymbol{a} + i\boldsymbol{b})e^{(\lambda+i\mu)t} \\
&= (\boldsymbol{a} + i\boldsymbol{b})e^{\lambda t}(\cos\mu t + i\sin\mu t) \\
&= e^{\lambda t}(\boldsymbol{a}\cos\mu t - \boldsymbol{b}\sin\mu t) + ie^{\lambda t}(\boldsymbol{a}\sin\mu t + \boldsymbol{b}\cos\mu t) \\
&= \boldsymbol{m}(t) + i\boldsymbol{n}(t)\,,
\end{aligned} \tag{4.23}$$

say. It can be shown that $\boldsymbol{m}$ and $\boldsymbol{n}$ are linearly independent solutions.

If a real $n \times n$ matrix $\boldsymbol{A}$ has two complex conjugate eigenvalues $r_1 = \lambda + i\mu$ and $r_2 = \lambda - i\mu$ with corresponding eigenvectors $\boldsymbol{\xi}^{(1)} = \boldsymbol{a} + i\boldsymbol{b}$ and $\boldsymbol{\xi}^{(2)} = \boldsymbol{a} - i\boldsymbol{b}$ and the other eigenvalues $r_3, \ldots, r_n$ are real and distinct with eigenvectors $\boldsymbol{\xi}^{(3)}, \ldots, \boldsymbol{\xi}^{(n)}$, then the general solution of (4.15) is

$$\boldsymbol{x} = c_1\boldsymbol{m}(t) + c_2\boldsymbol{n}(t) + c_3\boldsymbol{\xi}^{(3)}e^{r_3 t} + \cdots + c_n\boldsymbol{\xi}^{(n)}e^{r_n t}, \tag{4.24}$$

where $\boldsymbol{m}$ and $\boldsymbol{n}$ are given by (4.23).

**Example 4.13** Find a fundamental set of real-valued solutions of the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -\frac{1}{2} & 1 \\ -1 & -\frac{1}{2} \end{pmatrix} \boldsymbol{x}$$

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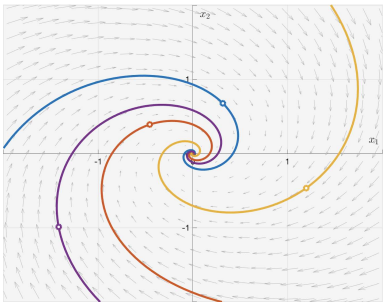


Figure 4.5: Phase portrait for example 4.13

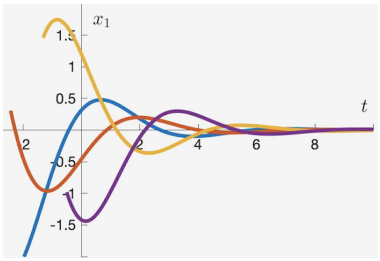



Figure 4.6: Plots of $x_1(t)$ for example 4.13.

### 4.4.3 Repeated eigenvalues

If $r = \rho$ is a $k$-fold root of the characteristic equation then $\rho$ is said to be an eigenvalue of multiplicity $k$ of the matrix $\boldsymbol{A}$. There are then two possibilities:

1. There are $k$ linearly independent eigenvectors $\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(k)}$ corresponding to the eigenvalue $\rho$. In this case the general solution is
$$\boldsymbol{x} = c_1 e^{\rho t}\boldsymbol{\xi}^{(1)} + \cdots + c_k e^{\rho t}\boldsymbol{\xi}^{(k)},$$
as expected.

2. There are fewer than $k$ such eigenvectors. In this case it is necessary to find other solutions of a different form. After finding solutions of the form
$$\boldsymbol{x} = \boldsymbol{\xi} e^{rt}$$
we then, following the procedure set out for second order linear equations, try solutions of the form
$$\boldsymbol{x} = \boldsymbol{\xi} t e^{rt} + \boldsymbol{\eta} e^{rt}$$
and then
$$\boldsymbol{x} = \boldsymbol{\theta} t^2 e^{rt} + \boldsymbol{\xi} t e^{rt} + \boldsymbol{\eta} e^{rt}$$
etc., until we find a complete set of linearly independent solutions.

**Example 4.14** Find the general solution of the autonomous system
$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \boldsymbol{x} \tag{4.25}$$
and draw a phase portrait for this system.

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## 4.5 Stability

*Reference: Boyce & DiPrima, Chapter 9 (but note some change in notation concerning the ordering of the eigenvalues)*

We seek a qualitative understanding of the behaviour of solutions of the homogeneous system of linear, first order ODEs

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x} \tag{4.26}$$

using geometrical methods. We concentrate on the case in which $\boldsymbol{A}$ is a $2 \times 2$ non-singular matrix (i.e. $|\boldsymbol{A}| \neq 0$) so that $\boldsymbol{x} = \boldsymbol{0}$ is the only **equilibrium solution** satisfying $d\boldsymbol{x}/dt = \boldsymbol{0}$.

Recall that in phase space[12], generic (non-equilibrium) solutions $\boldsymbol{x} = \boldsymbol{x}(t)$ are parametric curves that show the trajectory of a moving particle whose velocity $d\boldsymbol{x}/dt = \boldsymbol{A}\boldsymbol{x}$. Equilibrium solutions do not move and are often called **equilibrium points** (or fixed points, or critical points). A representative set of trajectories and equilibrium points in the phase space is called a **phase portrait**. Our aim is to characterise the differential equation according to the geometrical shape of its phase portrait. Several different cases need to be considered depending on the form of the eigenvalues of $\boldsymbol{A}$, $r_1$ and $r_2$.

[12] i.e. the $x_1x_2$-plane

### 4.5.1 Real and different eigenvalues with the same sign

The general solution of (4.26) is

$$\boldsymbol{x} = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t}, \tag{4.27}$$

where $r_1$ and $r_2$ are either both positive or both negative.

Consider first the case in which the eigenvalues are both negative with $r_1 < r_2 < 0$ and that the eigenvectors are as shown in figure 4.7. Then, (4.27) states that as $t \to \infty$, $\boldsymbol{x} \to \boldsymbol{0}$ regardless of the values of $c_1$ and $c_2$. Hence all solutions approach the equilibrium point at long times. If the solution starts at a point on the line through $\boldsymbol{\xi}^{(1)}$ then we know that $c_2 = 0$ and the solution remains on that line for all $t$, approaching the origin as $t \to \infty$. Similarly, if the initial conditions are such that a solution starts on the line through $\boldsymbol{\xi}^{(2)}$ then $c_1 = 0$ and solutions approach the origin along $\boldsymbol{\xi}^{(2)}$. More generally, it is helpful to write the solution (4.27) in the form

$$\boldsymbol{x} = e^{r_2 t}\left[c_1\boldsymbol{\xi}^{(1)}e^{(r_1 - r_2)t} + c_2\boldsymbol{\xi}^{(2)}\right]. \tag{4.28}$$

Then, since $r_1 - r_2 < 0$, the second term $c_2\boldsymbol{\xi}^{(2)}$ dominates the first for sufficiently large $t$, provided that $c_2 \neq 0$. Hence, as $t \to \infty$, the trajectory approaches the origin tangent to $\boldsymbol{\xi}^{(2)}$. Figure 4.7 shows an example phase portrait. Figure 4.8 shows examples of some solutions $x_1(t)$ ($x_2(t)$ are similar). This type of equilibrium point is called a **stable node** (aka nodal sink).

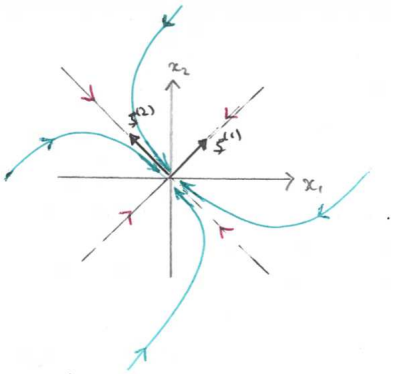



Figure 4.7: The phase portrait of a stable node with $r_1 < r_2 < 0$.

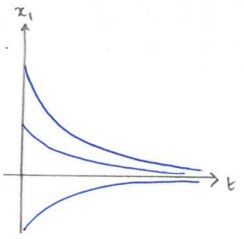



Figure 4.8: Examples of the solutions $x_1(t)$ for different initial conditions for the same case as Figure 4.7.

Now consider the case when the two eigenvalues are positive with $0 < r_1 < r_2$. In this case both $e^{r_1 t}$ and $e^{r_2 t}$ grow and so solutions move away from the origin. Since $r_1 - r_2 < 0$, the second term of (4.28) $c_2\boldsymbol{\xi}^{(2)}$ dominates the first for sufficiently large $t$, provided that $c_2 \neq 0$. Hence, as $t \to \infty$, the trajectory moves away from the origin parallel to $\boldsymbol{\xi}^{(2)}$. This type of equilibrium point is called a **unstable node** (aka nodal source).

From our previous examples, Example 4.12 is a stable node.

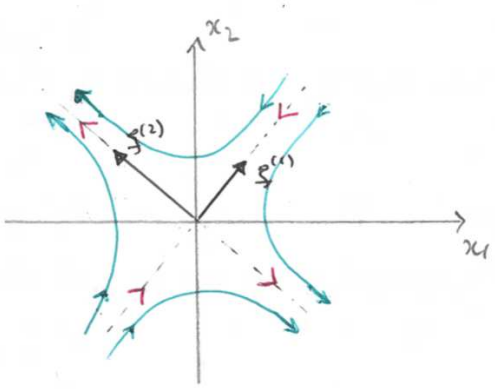



Figure 4.9: The phase portrait of a saddle point with $r_1 < 0 < r_2$.

### 4.5.2 Real and different eigenvalues with opposite sign

Now consider the case $r_1 < 0 < r_2$ and the solution

$$\boldsymbol{x} = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t}\,. \tag{4.29}$$

If the initial condition is on the line through $\boldsymbol{\xi}^{(1)}$ then $c_2 = 0$ and the solution remains on $\boldsymbol{\xi}^{(1)}$ for all time. Since $r_1 < 0$, $|\boldsymbol{x}| \to 0$ and the particle approaches the origin. Conversely, if the particle starts on the line through $\boldsymbol{\xi}^{(2)}$ then $c_1 = 0$ and the solution remains on $\boldsymbol{\xi}^{(2)}$ for all time. However, since $r_2 > 0$, $|\boldsymbol{x}| \to \infty$ and the solution moves away from the origin. For more generic initial conditions, eventually the solutions approach infinity parallel to the eigenvector $\boldsymbol{\xi}^{(2)}$ because the positive exponential term dominates for large $t$. An example of a phase portrait is shown in Figure 4.9.

Figure 4.10 shows examples of some solutions $x_1(t)$. For certain special initial conditions the positive exponential term is absent and $x_1 \to 0$ as $t \to 0$, while for all other initial conditions the positive exponential term eventually dominates and $x_1$ becomes unbounded. This type of equilibrium point is called a **saddle point**.

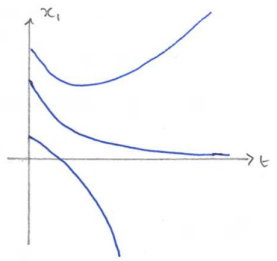



Figure 4.10: Examples of the solutions $x_1(t)$ for different initial conditions for the same case as in Figure 4.9.

From our previous examples, Example 4.11 is a saddle point.

### 4.5.3 Real and equal eigenvalues

Now we consider the case in which $r_1 = r_2 = r < 0$ (if $r > 0$ the trajectories in phase space are similar to what we describe below except the direction of motion is reversed). There are two subcases to consider depending on whether the repeated eigenvalue has two independent eigenvectors or just one.

1. If there are two independent eigenvectors then the solution has the form

$$\begin{aligned}\boldsymbol{x} = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} &= c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t} \\ &= e^{rt}(c_1\boldsymbol{\xi}^{(1)} + c_2\boldsymbol{\xi}^{(2)})\,.\end{aligned}$$

Note that the ratio of $x_1/x_2$ is independent of $t$ and so each trajectory is a straight line through the origin. Example trajectories are shown in Figure 4.11. This type of equilibrium point is called a **proper node** or **star point**. It is asymptotically stable (see the later definition 4.15) if $r < 0$ and unstable if $r > 0$.

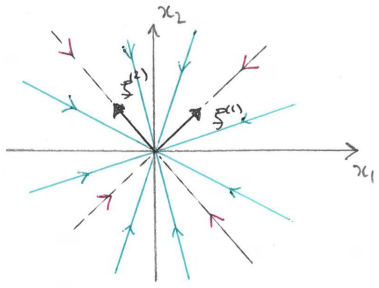



Figure 4.11: The phase portrait of an asymptotically stable proper node or star point with $r_1 = r_2 = r < 0$ and two independent eigenvectors.

2. When there is only one independent eigenvector the general solution takes the form (see §4.4.3)

$$\begin{aligned} \boldsymbol{x} &= c_1\boldsymbol{\xi}e^{rt} + c_2(\boldsymbol{\xi}te^{rt} + \boldsymbol{\eta}e^{rt})\,. \\ &= [(c_1\boldsymbol{\xi} + c_2\boldsymbol{\eta}) + c_2\boldsymbol{\xi}t]e^{rt}\,, \\ &= \boldsymbol{y}e^{rt}, \end{aligned} \tag{4.30}$$

say. As $t \to \infty$ the dominant term is $c_2\boldsymbol{\xi}te^{rt}$ and so at large times every trajectory approaches the origin tangent to $\boldsymbol{\xi}$. (Note that this is true even if $c_2 = 0$ because then the solution is $c_1\boldsymbol{\xi}e^{rt}$.) Similarly, if we also consider $t \to -\infty$ then again the term $c_2\boldsymbol{\xi}te^{rt}$ dominates. Hence the trajectory is parallel to the vector $\boldsymbol{\xi}$ far into the future and far into the past. An example of the phase portrait is shown in figure 4.12.

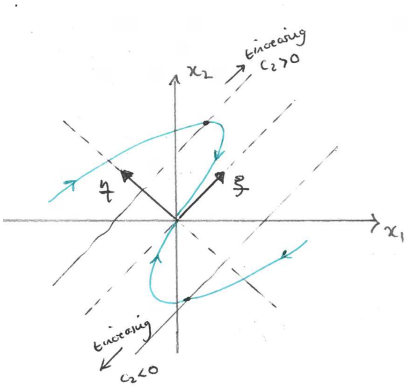



Figure 4.12: The phase portrait of an asymptotically stable improper node or degenerate point with $r_1 = r_2 = r < 0$ and only one independent eigenvector.

This type of equilibrium point is called a **improper node** or **degenerate node**. As for the proper node, it is asymptotically stable if $r < 0$ and unstable if $r > 0$.

From our previous examples, Example 4.14 is an unstable improper node.

### 4.5.4 Complex conjugate eigenvalues

First we consider the case in which the eigenvalues are $\lambda \pm i\mu$ (where $\lambda$ and $\mu$ are real) with $\lambda \neq 0$ and $\mu > 0$. These systems typically take the form

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} \lambda & \mu \\ -\mu & \lambda \end{pmatrix} \boldsymbol{x} \tag{4.31}$$

which is equivalent to

$$x_1' = \lambda x_1 + \mu x_2, \quad x_2' = -\mu x_1 + \lambda x_2\,. \tag{4.32}$$

We can simplify this system by changing to polar coordinates $r$ and $\theta$ given by

$$r^2 = x_1^2 + x_2^2, \quad \tan\theta = \frac{x_2}{x_1}\,.$$

Differentiating these expressions with respect to $t$ gives

$$rr' = x_1x_1' + x_2x_2', \quad \theta' \sec^2\theta = \frac{x_1x_2' - x_2x_1'}{x_1^2}$$

and substituting for $x_1'$ and $x_2'$ from (4.32) gives

$$r' = \lambda r, \quad \theta' = -\mu\,,$$

since $\sec^2\theta = 1 + \tan^2\theta$. Integrating gives

$$r = ce^{\lambda t}, \quad \theta = -\mu t + \theta_0\,,$$

where $c$ is a constant and $\theta_0 = \theta(t = 0)$. Thus we have parametric equations of the trajectories in polar coordinates. Since $\mu > 0$, $\theta$ decreases as $t$ increases and the motion is clockwise in the $r\theta$-plane.

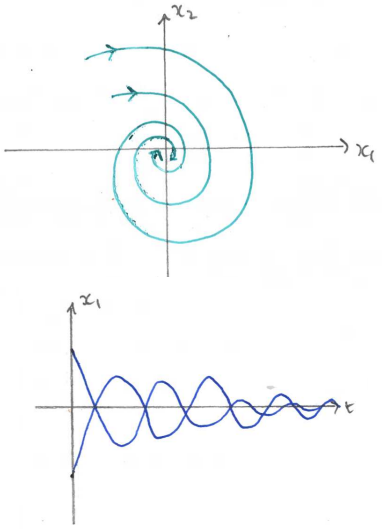



Figure 4.13: The phase portrait of an asymptotically stable spiral sink (top) and examples of the solutions $x_1(t)$ for different initial conditions (bottom).

As $t \to \infty$, $r \to 0$ if $\lambda < 0$ and $r \to \infty$ if $\lambda > 0$. Hence the trajectories are spirals which approach ($\lambda < 0$) or move away from ($\lambda > 0$) the origin. This type of equilibrium point is called an asymptotically stable **spiral sink** when $\lambda < 0$ (aka stable focus) or an unstable **spiral source** when $\lambda > 0$ (aka unstable focus). Examples of phase portraits and the solutions $x_1(t)$ are shown in figures 4.13 and figures 4.14.

From our previous examples, Example 4.13 is a spiral sink.

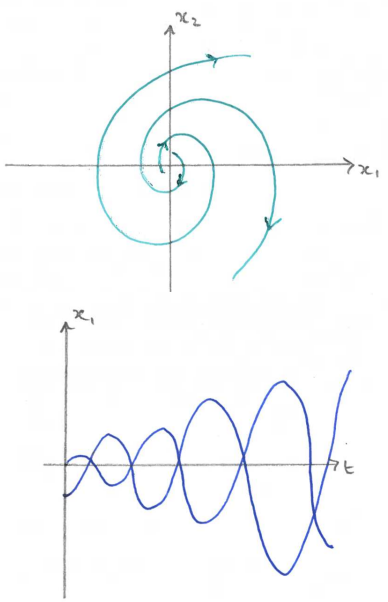



Figure 4.14: The phase portrait of an unstable stable spiral source (top) and examples of the solutions $x_1(t)$ for different initial conditions (bottom).

**Pure imaginary eigenvalues**

Note that when we have pure imaginary eigenvalues (i.e. $\lambda = 0$) the same analysis as above leads to

$$r = c, \quad \theta = -\mu t + \theta_0 \,,$$

where $c$ and $\theta_0$ are constants. Hence the trajectories are circles, with centre the origin. They are traversed clockwise if $\mu > 0$ and anti-clockwise if $\mu < 0$. The solutions are periodic with period $2\pi/\mu$. The equilibrium point is referred to as a **centre** and it is stable. An example phase portrait and some solutions $x_1(t)$ are shown in figure 4.15.

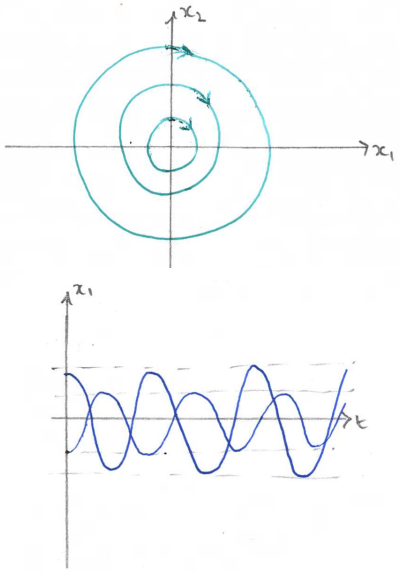



Figure 4.15: The phase portrait of a stable centre (top) and examples of the solutions $x_1(t)$ for different initial conditions (bottom).

In the above discussion, we have referred to stable and unstable equilibrium points based on the geometrical structure of their phase portrait. To be more precise, we have the following definition:

**Definition 4.15** For the autonomous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x}\,, \tag{4.33}$$

where $\boldsymbol{A}$ is a constant $n \times n$ matrix, any points where $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ are **equilibrium points**.

An equilibrium point $\boldsymbol{x}^*$ is said to be **stable** if, given any $\epsilon > 0$, there exists a $\delta > 0$ such that every solution $\boldsymbol{x}(t)$ of system (4.33) that at $t = 0$ satisfies

$$|\boldsymbol{x}(0) - \boldsymbol{x}^*| < \delta,$$

exists for all $t > 0$ and satisfies

$$|\boldsymbol{x}(t) - \boldsymbol{x}^*| < \epsilon,$$

for all $t \geq 0$. An equilibrium point that is not stable is said to be **unstable**.

An equilibrium point is said to be **asymptotically stable** if it is stable and if there exists a $\delta_0$, with $0 < \delta_0 < \delta$, such that if a solution $\boldsymbol{x}(t)$ satisfies

$$|\boldsymbol{x}(0) - \boldsymbol{x}^*| < \delta_0,$$

then

$$\lim_{t\to\infty} \boldsymbol{x}(t) = \boldsymbol{x}^*\,. \tag{4.34}$$

Informally, stable solutions are those that start sufficiently close to $\boldsymbol{x}^*$ (within $\delta$) and stay close (within $\epsilon$). For asymptotic stability, solutions must not only stay close, but must eventually approach $\boldsymbol{x}^*$ as $t \to \infty$.

The above ideas of stability will be useful in modules such as MTH3011 (Nonlinear Systems and Control) and MTH3039 (Computational Nonlinear Dynamics) where one encounters **nonlinear systems**

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{f}(\boldsymbol{x})$$

which can be approximated by an **almost linear system**

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(\boldsymbol{x})$$

in the neighbourhood of the equilibrium point. One is then interested in whether the trajectories of the linear system are good approximations to the behaviour of the nonlinear system. We will not pursue this further in this module.

## 4.6 Fundamental matrix

**Definition 4.16** A set of $n$ linearly independent solutions $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ of the system of $n$ first order linear equations

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} \tag{4.35}$$

on the interval $I : \alpha < t < \beta$ is said to be a **fundamental set of solutions** on that interval.

**Definition 4.17** A square matrix function $\boldsymbol{\Psi}(t)$

$$\boldsymbol{\Psi}(t) = \begin{pmatrix} x_1^{(1)}(t) & \ldots & x_1^{(n)}(t) \\ \vdots & & \vdots \\ x_n^{(1)}(t) & \ldots & x_n^{(n)}(t) \end{pmatrix} \tag{4.36}$$

whose columns are a fundamental set of solutions of system (4.35) is called a **fundamental matrix** of that system. Note that a fundamental matrix is nonsingular since its columns are linearly independent vectors.

**Example 4.18** Find the fundamental matrix for the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \boldsymbol{x}$$

### 4.6.1 General solution in terms of the fundamental matrix

When written in terms of the fundamental matrix $\mathbf{\Psi}(t)$, the general solution

$$\boldsymbol{x} = c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_n\boldsymbol{x}^{(n)}(t)$$

of (4.35) becomes

$$\boldsymbol{x} = \mathbf{\Psi}(t)\boldsymbol{c}\,. \tag{4.37}$$

Here $\boldsymbol{c} = (c_1, \ldots, c_n)^T$ is the vector of arbitrary constants. For the initial value problem comprising the differential equation (4.35) and the initial conditions

$$\boldsymbol{x}(t_0) = \boldsymbol{k}\,, \tag{4.38}$$

where $t_0$ is a given point in the interval $I : \alpha < t < \beta$ and $\boldsymbol{k}$ is a given initial vector, we must choose $\boldsymbol{c}$ so that

$$\boldsymbol{x}(t_0) = \mathbf{\Psi}(t_0)\boldsymbol{c}\,.$$

Since $\mathbf{\Psi}(t_0)$ is nonsingular we have

$$\boldsymbol{c} = \mathbf{\Psi}(t_0)^{-1}\boldsymbol{x}(t_0)$$

and the solution of the initial value problem is then, from (4.37),

$$\boldsymbol{x} = \mathbf{\Psi}(t)\mathbf{\Psi}(t_0)^{-1}\boldsymbol{x}(t_0)\,. \tag{4.39}$$

Since each column of the fundamental matrix $\mathbf{\Psi}$ is a solution of (4.35), it follows that $\mathbf{\Psi}$ satisfies the matrix differential equation

$$\frac{d\mathbf{\Psi}}{dt} = \boldsymbol{P}(t)\mathbf{\Psi}\,. \tag{4.40}$$

**Activity 4.19** Confirm (4.40) by comparing the $j^{th}$ columns of each side.

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### 4.6.2 Special form of the fundamental matrix

**Theorem 4.20** Let

$$\boldsymbol{e}^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \boldsymbol{e}^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots \quad \boldsymbol{e}^{(n)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}, \tag{4.41}$$

and let $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ be the solutions of system (4.35) that satisfy the initial conditions

$$\boldsymbol{x}^{(1)}(t_0) = \boldsymbol{e}^{(1)}, \quad \ldots, \quad \boldsymbol{x}^{(n)}(t_0) = \boldsymbol{e}^{(n)}, \tag{4.42}$$

where $t_0$ is any point in the interval $I : \alpha < t < \beta$. Then $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ form a fundamental set of solutions of system (4.35).

Sometimes it is useful to use the **special fundamental matrix**, that we will denote by $\boldsymbol{\Phi}(t)$, whose columns are the vectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ defined in Theorem 4.20. Then, at $t = t_0$ we have

$$\boldsymbol{\Phi}(t_0) = \begin{pmatrix} 1 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & 1 \end{pmatrix} = \boldsymbol{I}. \tag{4.43}$$

Now note that $\boldsymbol{\Phi}^{-1}(t_0) = \boldsymbol{I}$ and so from (4.39) we have

$$\boldsymbol{x} = \boldsymbol{\Phi}(t)\boldsymbol{x}(t_0). \tag{4.44}$$

Expression (4.44) represents a transformation of the initial condition $\boldsymbol{x}(t_0)$ into the solution $\boldsymbol{x}(t)$ at arbitrary time $t$. The advantage of (4.44) is that once the special fundamental matrix $\boldsymbol{\Phi}$ has been found, we can find the solution of a matrix system with many different initial conditions simply through matrix multiplication. Comparing (4.39) and (4.44) we see that

$$\boldsymbol{\Phi}(t) = \boldsymbol{\Psi}(t)\boldsymbol{\Psi}^{-1}(t_0). \tag{4.45}$$

**Example 4.21** Find the special fundamental matrix $\boldsymbol{\Phi}(t)$ with $\boldsymbol{\Phi}(0) = \boldsymbol{I}$ for the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \boldsymbol{x}\,.$$

.

## 4.7 Nonhomogeneous Linear Systems

We now consider the nonhomogeneous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} + \boldsymbol{g}(t)\,, \tag{4.46}$$

where the $n \times n$ matrix $\boldsymbol{P}(t)$ and the column vector $\boldsymbol{g}(t)$ are continuous on the interval $I : \alpha < t < \beta$. The general solution of (4.46) can be written in the form

$$\boldsymbol{x} = c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_n\boldsymbol{x}^{(n)}(t) + \boldsymbol{w}(t)\,, \tag{4.47}$$

where $\boldsymbol{x}^{(1)}(t), \ldots, \boldsymbol{x}^{(n)}(t)$ is a fundamental set of solutions of the corresponding homogeneous problem

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} \tag{4.48}$$

and $\boldsymbol{w}(t)$ is a particular solution of the nonhomogeneous system (4.46). We will consider several methods of finding the particular solution $\boldsymbol{w}$.

### 4.7.1 Method of Undetermined Coefficients

This method is useful if the coefficient matrix $\boldsymbol{P}$ is a constant matrix ($= \boldsymbol{A}$, say) and if the components of $\boldsymbol{g}$ are polynomials, exponentials, sinusoidal functions or sums or products of these. The method follows that for second order ODEs, where we search for a particular integral of the same form as $\boldsymbol{g}$. One main difference is when $\boldsymbol{g}$ has the form $\boldsymbol{u}e^{\lambda t}$ and $\lambda$ is a simple root of the characteristic equation. In this case, we might expect that we should seek solutions of the form $\boldsymbol{a}te^{\lambda t}$. However, it is necessary to use $\boldsymbol{a}te^{\lambda t} + \boldsymbol{b}e^{\lambda t}$. Then $\boldsymbol{a}$ and $\boldsymbol{b}$ are determined by substitution.

**Example 4.22** Use the method of undetermined coefficients to find the general solution of

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\boldsymbol{x} + \begin{pmatrix} 2e^{-t} \\ 3t \end{pmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)$$

.

.

### 4.7.2 Method of Diagonalisation

Consider the nonhomogeneous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)\,, \tag{4.49}$$

where the $n \times n$ diagonalisable matrix $\boldsymbol{A}$ is a constant matrix. The method proceeds by defining a new variable $\boldsymbol{y}$ by

$$\boldsymbol{x} = \boldsymbol{T}\boldsymbol{y}\,,$$

where $\boldsymbol{T}$ is a matrix whose columns are the (constant) eigenvectors of $\boldsymbol{A}$. Then by substituting into (4.49) we obtain

$$\boldsymbol{T}\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{T}\boldsymbol{y} + \boldsymbol{g}(t)\,.$$

Now we multiply by $\boldsymbol{T}^{-1}$ and find

$$\boldsymbol{y}' = (\boldsymbol{T}^{-1}\boldsymbol{A}\boldsymbol{T})\boldsymbol{y} + \boldsymbol{T}^{-1}\boldsymbol{g}(t) = \boldsymbol{D}\boldsymbol{y} + \boldsymbol{h}(t)\,, \tag{4.50}$$

where $\boldsymbol{D}$ is the diagonal matrix whose diagonal entries are the eigenvalues $r_1, \ldots, r_n$ arranged in the same order as the corresponding eigenvectors $\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(n)}$ that form the columns of $\boldsymbol{T}$. System (4.50) is then a system of uncoupled linear ODEs for $y_1(t), \ldots, y_n(t)$, which can be solved by the methods of § 2.

**Example 4.23** Find the general solution of the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\boldsymbol{x} + \begin{pmatrix} 2e^{-t} \\ 3t \end{pmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)$$

using the method of diagonalisation.

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### 4.7.3 Variation of Parameters

Now we consider the more general nonhomogeneous problem

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} + \boldsymbol{g}(t), \tag{4.51}$$

with $\boldsymbol{P}(t)$ and $\boldsymbol{g}(t)$ continuous on the interval $I : \alpha < t < \beta$, in which the matrix $\boldsymbol{P}$ is not constant or not diagonalisable. We assume that the fundamental matrix for the corresponding homogeneous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} \tag{4.52}$$

has been found and we use the method of variation of parameters to find a particular solution (and hence the general solution) of (4.51).

We know that the general solution of the homogeneous problem (4.52) is $\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{c}$ (see (4.37)) and so, as we did for second order ODEs in §3.6, we seek a solution of the nonhomogeneous problem by replacing $\boldsymbol{c}$ by an unknown $\boldsymbol{u}(t)$, i.e. we set

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{u}(t)\,,$$

where $\boldsymbol{u}(t)$ is to be found. Differentiating the product and substituting into (4.51) gives

$$\begin{aligned} \boldsymbol{\Psi}'(t)\boldsymbol{u}(t) + \boldsymbol{\Psi}(t)\boldsymbol{u}'(t) &= \boldsymbol{P}(t)\boldsymbol{\Psi}(t)\boldsymbol{u}(t) + \boldsymbol{g}(t)\,, \\ \Rightarrow \boldsymbol{\Psi}(t)\boldsymbol{u}'(t) &= \boldsymbol{g}(t)\,, \end{aligned}$$

because the fundamental matrix satisfies $\boldsymbol{\Psi}'(t) = \boldsymbol{P}(t)\boldsymbol{\Psi}(t)$ (see (4.40)). Now, $\boldsymbol{\Psi}(t)$ is nonsingular on any interval where $\boldsymbol{P}$ is continuous, so $\boldsymbol{\Psi}^{-1}(t)$ exists and we have

$$\begin{aligned} \boldsymbol{u}'(t) &= \boldsymbol{\Psi}^{-1}(t)\boldsymbol{g}(t) \\ \Rightarrow \boldsymbol{u}(t) &= \int \boldsymbol{\Psi}^{-1}(t)\boldsymbol{g}(t)dt + \boldsymbol{c}\,, \end{aligned} \tag{4.53}$$

where the constant vector $\boldsymbol{c}$ is arbitrary. Hence we find

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{c} + \boldsymbol{\Psi}(t)\int \boldsymbol{\Psi}^{-1}(t)\boldsymbol{g}(t)dt\,. \tag{4.54}$$

Note that any initial condition can be satisfied by an appropriate choice for the arbitrary constant $\boldsymbol{c}$ and so (4.54) is the general solution. The first term on the rhs is the general solution of the homogeneous system (4.52) and the second term is a particular integral of (4.51).

For the initial conditions

$$\boldsymbol{x}(t_0) = \boldsymbol{k}$$

we can choose the particular solution in (4.54) as the one that is zero when $t = t_0$. This can be achieved by writing the general solution (4.54) in the form

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{c} + \boldsymbol{\Psi}(t)\int_{t_0}^{t} \boldsymbol{\Psi}^{-1}(s)\boldsymbol{g}(s)ds\,. \tag{4.55}$$

and taking

$$\boldsymbol{c} = \boldsymbol{\Psi}^{-1}(t_0)\boldsymbol{x}(t_0)$$

so that the solution of the initial value problem is

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{\Psi}^{-1}(t_0)\boldsymbol{x}(t_0) + \boldsymbol{\Psi}(t)\int_{t_0}^{t} \boldsymbol{\Psi}^{-1}(s)\boldsymbol{g}(s)ds\,. \tag{4.56}$$

Finally, note that if we use the special fundamental matrix $\boldsymbol{\Phi}(t)$ with $\boldsymbol{\Phi}(t_0) = \boldsymbol{I}$ then we have the simpler form

$$\boldsymbol{x} = \boldsymbol{\Phi}(t)\boldsymbol{x}(t_0) + \boldsymbol{\Phi}(t)\int_{t_0}^{t} \boldsymbol{\Phi}^{-1}(s)\boldsymbol{g}(s)ds\,. \tag{4.57}$$

**Example 4.24** Find the general solution of the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\boldsymbol{x} + \begin{pmatrix} 2e^{-t} \\ 3t \end{pmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)$$

using the method of variation of parameters.

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